

Learning Resources Guide

Synced with Your AI Data Engineer Roadmap v2

Every roadmap topic → Best free website + Udemy Business course — Validated May 2026

What changed in v2

- DataLemur and Interview Query removed — replaced by DataDriven.io (100% free, 1,400+ problems, no credit card)
- Docker resources added: docs.docker.com/get-started
- Terraform resources added: developer.hashicorp.com/terraform/tutorials
- Apache Iceberg resources added: iceberg.apache.org/docs + datalakehousehub.com
- Airbyte and dlt resources added: docs.airbyte.com + dlthub.com/docs
- Data Observability resources added: Great Expectations + Soda Core + Monte Carlo blog
- Git / GitHub resources added: skills.github.com (interactive, free)

Blue bar	= Learn only
Green bar	= Practice only
Amber bar	= Learn + Practice or Udemy Business course

MONTH 1 — Foundations (Weeks 1–4)

W1 — Git & GitHub (NEW)

Resource	URL / Search term	What you learn / use it for
GitHub Skills Learn	skills.github.com	Free interactive courses: intro to GitHub, branches, pull requests — runs in browser, ~2 hrs
GitHub Docs: Get Started Learn	docs.github.com/get-started	Official guide: create a repo, first commit, branching, pull requests — free reference
Git Tutorial (Atlassian) Learn	atlassian.com/git/tutorials	Free in-depth guide to Git workflows, rebasing, hooks — well illustrated

W1–2 — Python (data focus)

Resource	URL / Search term	What you learn / use it for
Kaggle Python Course Learn	kaggle.com/learn/python	Free: syntax, loops, functions, list comprehensions, file I/O — interactive notebooks
Kaggle Pandas Course Learn+Practice	kaggle.com/learn/pandas	Free: DataFrames, groupby, merge, pivot, apply — the exact patterns used in pipelines
Real Python Learn	realpython.com	Free articles on Pydantic, asyncio, Pytest, type hints, logging — bookmark and read daily
Python Bootcamp (Jose Portilla) Udemy Business	Search: "python bootcamp portilla"	Comprehensive Python: OOP, decorators, virtual envs, argparse — well-structured foundation
Automate the Boring Stuff Udemy Business	Search: "automate boring stuff python"	Deep Python fluency: file I/O, APIs, error handling — ideal for scripting pipelines

W1–2 — Python practice

Resource	URL / Search term	What you learn / use it for
HackerRank Python Practice	hackerrank.com/domains/python	Graded from easy → hard — used by Deloitte & many firms for screening
Exercism Python Track Practice	exercism.org/tracks/python	Mentor-reviewed exercises — great for writing clean, idiomatic Python

W3 — Advanced SQL

Resource	URL / Search term	What you learn / use it for
Mode SQL Tutorial Learn	mode.com/sql-tutorial	Free 6-hr course — window functions, CTEs, subqueries, performance. Live SQL in browser
SQLZoo Learn+Practice	sqlzoo.net	Interactive lessons: COUNT → window functions → self-joins. Small, focused exercises
The Complete SQL Bootcamp Udemy Business	Search: "complete sql bootcamp"	PostgreSQL end-to-end — window functions, CTEs, indexing, optimization

W3 — SQL practice (daily)

Resource	URL / Search term	What you learn / use it for
DataDriven.io — SQL Practice	datadriven.io	100% free, 1,400+ real DE interview problems. No credit card. No paywall. No expiry. USE THIS DAILY
LeetCode SQL 50 Practice	leetcode.com/studyplan/top-sql-50	Curated 50 SQL problems at DE interview difficulty — free, widely referenced in job prep

W4 — Cloud (pick AWS or GCP)

Resource	URL / Search term	What you learn / use it for
AWS Skill Builder Learn	skillbuilder.aws	Free official courses: S3, Glue, Redshift, Lambda, IAM — create a free account
AWS Free Tier Learn+Practice	aws.amazon.com/free	12 months free: S3, Lambda, RDS, Glue — build your W4 pipeline at zero cost
Google Cloud Skills Boost Learn+Practice	cloudskillsboost.google	Free labs: BigQuery, Dataflow, Pub/Sub — earn shareable badges, \$300 new-account credit
BigQuery Sandbox Practice	cloud.google.com/bigquery/docs/sandbox	Free tier, no card — run SQL on public datasets, 10 GB storage free
AWS Certified Data Engineer Associate Udemy Business	Search: "aws data engineer associate"	S3, Glue, Redshift, Kinesis, Athena end-to-end — exam prep + structured learning
GCP Professional Data Engineer Udemy Business	Search: "google cloud professional data engineer"	BigQuery, Dataflow, Pub/Sub, Vertex AI — solid GCP cert prep

MONTH 2 — Core Data Engineering (Weeks 5–10)

W5 — Docker & Containers (NEW)

Resource	URL / Search term	What you learn / use it for
Docker Get Started Learn	docs.docker.com/get-started	Official free tutorial: containers, images, compose — ~4 hrs, browser-readable
Docker Compose Docs Learn+Practice	docs.docker.com/compose	Spin up multi-service stacks — use this to run Postgres + pgvector, Kafka locally
Play with Docker Practice	labs.play-with-docker.com	Free browser-based Docker playground — no install needed, run commands live
Docker and Kubernetes: The Complete Guide Udemy Business	Search: "docker kubernetes complete guide"	Deep Docker + Compose for engineers — covers everything you need for DE environments

W6 — Terraform (NEW)

Resource	URL / Search term	What you learn / use it for
Terraform Tutorials (HashiCorp) Learn+Practice	developer.hashicorp.com/terraform/tutorials	Free interactive browser labs — provision AWS/GCP resources in Terraform, ~3 hrs
Terraform Official Docs Learn	developer.hashicorp.com/terraform/docs	Complete reference — providers, resources, state management
Terraform: From Beginner to Advanced Udemy Business	Search: "terraform beginner advanced"	Hands-on Terraform for AWS/GCP — covers all patterns expected in DE roles

W6 — Airflow

Resource	URL / Search term	What you learn / use it for
Astronomer Academy (Airflow 3) Learn	academy.astronomer.io	Free courses by the Airflow experts: DAG authoring, sensors, dynamic tasks, RAG with Airflow
DataTalks.Club DE Zoomcamp Learn+Practice	github.com/DataTalksClub/data-engineering-zoomcamp	Free 9-week cohort (Jan 2026) — Airflow module + hands-on pipeline projects
Astro CLI (local Airflow) Practice	docs.astronomer.io/astro/cli/install-cli	Free local Airflow — run DAGs on your laptop in minutes
Complete Hands-On Apache Airflow Udemy Business	Search: "marc lamberti airflow"	Marc Lamberti's course — the most recommended Airflow course on r/dataengineering

W7 — dbt

Resource	URL / Search term	What you learn / use it for
dbt Fundamentals Learn	courses.getdbt.com/courses/fundamentals	Official free 6-hr course: models, tests, documentation, sources, ref() — start here

dbt Learn — Advanced dbt Learn	courses.getdbt.com/courses/advanced-dbt	Incremental models, snapshots, testing — free official course
Jaffle Shop (practice project) Practice	github.com/dbt-labs/jaffle_shop	The canonical dbt learning project — build a complete dbt project from scratch
dbt Cloud Free Tier Practice	cloud.getdbt.com	Free developer account — run dbt against BigQuery/Snowflake in browser
dbt Bootcamp: Zero to Hero Udemy Business	Search: "dbt bootcamp zero to hero"	dbt Core + dbt Cloud, BigQuery & Snowflake, end-to-end project

W7 — Apache Iceberg & Delta Lake (NEW)

Resource	URL / Search term	What you learn / use it for
Apache Iceberg Docs Learn	iceberg.apache.org/docs	Official documentation: table format, schema evolution, time travel, partitioning
Iceberg & Lakehouse Hub Learn	datalakehousehub.com	Free 2025-26 guide to Iceberg with beginner tutorials, local setup walkthroughs
Databricks Delta Lake (on Community) Learn+Practice	docs.delta.io/latest/index.html	Delta Lake docs + practice on Databricks Community Edition (free Spark environment)

W8 — Airbyte & dlt (NEW)

Resource	URL / Search term	What you learn / use it for
Airbyte Docs Learn+Practice	docs.airbyte.com	Full docs: quickstart, connectors, deploying via Docker Compose locally — free/open source
dlt (data load tool) Docs Learn+Practice	dlthub.com/docs	pip install dlt — quickstart in 10 min, REST API & database source tutorials — free
Data Engineering with Airbyte, dbt & Airflow Udemy Business	Search: "airbyte dbt airflow pipeline"	Modern stack course: Airbyte ingestion + dbt transform + Airflow orchestration

W9 — Apache Spark (PySpark)

Resource	URL / Search term	What you learn / use it for
Databricks Community Edition Learn+Practice	community.cloud.databricks.com	Free Spark + Delta Lake notebooks in the cloud — the single best free Spark environment
TrendyTech (Sumit Mittal) — YouTube Learn	youtube.com/@trendytech	Deep PySpark tutorials: DataFrames, joins, window functions, optimisation — DE-focused
Apache Spark with Python — PySpark Udemy Business	Search: "apache spark pyspark frank kane"	Frank Kane's course — practical PySpark, DataFrames, real datasets

W10 — Kafka

Resource	URL / Search term	What you learn / use it for
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Confluent Developer Learn	developer.confluent.io/learn-kafka/	Free Kafka 101: topics, partitions, offsets, consumer groups — official Confluent course
Kafka on Docker (Conduktor guide) Practice	conduktor.io/kafka/how-to-start-kafka-with-docker	Free guide to running Kafka locally on Docker — Python producers & consumers
Confluent Cloud Free Tier Practice	confluent.cloud	Free managed Kafka account — \$400 credit, no local setup needed
Apache Kafka Series — Learn Kafka for Beginners Udemy Business	Search: " kafka beginners stephane maarek "	Stephane Maarek's course — widely regarded as the best Kafka resource

MONTH 3 — AI Data Engineering + Portfolio (Weeks 11–14)

W11 — Data Observability (NEW)

Resource	URL / Search term	What you learn / use it for
Great Expectations Docs Learn+Practice	docs.greatexpectations.io	Free data quality framework — quickstart in 15 min, add to your pipeline, pip install
Soda Core Docs Learn+Practice	docs.soda.io	Free open-source data quality — pip install, works with BigQuery/Snowflake/Postgres
Monte Carlo Blog Learn	montecarlodata.com/blog	Free articles on all 5 observability pillars, data lineage, freshness monitoring
OpenLineage Docs Learn	openlineage.io	Free open standard for data lineage — integrates with Airflow, Spark, dbt

W12 — ML Pipeline Infrastructure

Resource	URL / Search term	What you learn / use it for
Kaggle Feature Engineering Course Learn	kaggle.com/learn/feature-engineering	Free: mutual information, encoding, feature creation — hands-on notebooks
MLflow Official Docs + Quickstart Learn+Practice	mlflow.org/docs/latest/quickstart.html	Free, runs locally — track experiments, log models, compare runs — 30 min to first run
DVC Docs Learn+Practice	dvc.org/doc/start	Free, Git-based data versioning — version datasets and ML models like code
DataTalks.Club MLOps Zoomcamp Learn+Practice	github.com/DataTalksClub/mlops-zoomcamp	Free 9-week MLOps: MLflow, Prefect, Docker, monitoring — ideal complement
MLOps Engineering on AWS Udemy Business	Search: "mlops engineering aws"	End-to-end ML pipelines: SageMaker, MLflow, feature stores, model serving

W12 — LLM / RAG / Vector DBs

Resource	URL / Search term	What you learn / use it for
LangChain Python Docs Learn+Practice	python.langchain.com/docs	Official docs with tutorials: document loaders, text splitters, embedding chains, RAG
OpenAI Cookbook Learn+Practice	cookbook.openai.com	Free tutorials: embeddings, RAG, data extraction — copy-paste Python notebooks
Pinecone Learning Center Learn	pinecone.io/learn	Free vector DB course: embeddings, semantic search, RAG architecture — 3 hrs with code
Microsoft Gen AI for Beginners Learn	github.com/microsoft/generative-ai-for-beginners	Free 18-lesson course: RAG (lesson 15), vector DBs, LLM pipelines — Python throughout
DataTalks.Club LLM Zoomcamp Learn+Practice	github.com/DataTalksClub/llm-zoomcamp	Free cohort on LLM engineering, RAG systems, evaluation — same community as DE Zoomcamp
LangChain — Chat with Your Data	Search: "langchain chat data harrison chase"	Harrison Chase (LangChain creator) teaches RAG, document Q&A, vector stores

Udemy Business**Vector Databases: Embeddings to Applications**
Udemy Business

Search: "vector databases embeddings"

Covers pgvector, Pinecone, Weaviate — hands-on embedding pipeline projects

W12 — Vector DB practice

Resource	URL / Search term	What you learn / use it for
pgvector on Docker Practice	github.com/pgvector/pgvector	Open-source Postgres vector extension — run locally via Docker, used in capstone
Pinecone Free Tier Practice	app.pinecone.io	Free serverless index, no card — build RAG pipelines, enough for capstone

W13 — Capstone: Streamlit + Data Quality

Resource	URL / Search term	What you learn / use it for
Streamlit Docs Learn+Practice	docs.streamlit.io	Official docs + quickstart — build a data app in Python in minutes
Streamlit Community Cloud Practice	streamlit.io/cloud	Deploy your capstone for free — public URL for your resume and LinkedIn
Kaggle HR Dataset Practice	kaggle.com/datasets search: ibm hr attrition	IBM HR attrition dataset — 1470 rows, perfect for ML + dbt + embedding pipeline

W14 — Certifications

Resource	URL / Search term	What you learn / use it for
AWS Certified Data Engineer – Associate Learn	aws.amazon.com/certification/certified-data-engineer-associate	Official exam guide, free practice questions, sample exam — primary cert target
AWS Skill Builder DE Exam Prep Learn	skillbuilder.aws search: data engineer	Free official exam prep from AWS — use before paying for practice exams
GCP Professional Data Engineer Learn	cloud.google.com/learn/certification/data-engineer	Exam guide + free sample questions — strong if Deloitte work is GCP-aligned
AWS DE Associate Practice Exams Udemy Business	Search: "aws certified data engineer associate practice"	6 full practice exams — use in final 2 weeks before sitting the exam
GCP Professional DE Practice Exams Udemy Business	Search: "gcp professional data engineer practice exams"	Full mock exams — widely used for GCP cert prep

W14 — Interview practice

Resource	URL / Search term	What you learn / use it for
DataDriven.io — DataDriven 75 Practice	datadriven.io/playbooks/datadriven-75	75 curated DE interview problems, real code execution, AI grading, company tags — 100% free
DataDriven.io — Mock Interview Practice	datadriven.io/interview	AI mock interviews with hire/no-hire verdicts — free, simulates real onsite format

Ongoing — Communities, YouTube & Newsletters

Communities — join from Week 1

Resource	Type	Time	Link / Platform
DataTalks.Club Slack	Community	Ongoing	datatalks.club/slack.html
dbt Community Slack	Community	Ongoing	community.getdbt.com
Apache Airflow Slack	Community	Ongoing	apache-airflow.slack.com
r/dataengineering	Reddit	Ongoing	reddit.com/r/dataengineering
LinkedIn #DataEngineering	Social	Ongoing	linkedin.com — follow hashtag

YouTube channels — subscribe from Day 1

Resource	Type	Time	Link / Platform
Darshil Parmar	YouTube	Ongoing	youtube.com/@darshilparmar
Seattle Data Guy	YouTube	Ongoing	youtube.com/@SeattleDataGuy
TrendyTech (Spark focus)	YouTube	Ongoing	youtube.com/@trendytech
Andreas Kretz (architectures)	YouTube	Ongoing	youtube.com/@andreaskayy
Data with Zach (Zach Wilson)	YouTube	Ongoing	youtube.com/@datawithzach
Matt Palmer (dbt focus)	YouTube	Ongoing	youtube.com/@mattpalmer

Newsletters

Resource	Type	Time	Link / Platform
Data Engineering Weekly	Free Newsletter	Weekly	dataengineeringweekly.com
DataTalks.Club Newsletter	Free Newsletter	Weekly	datatalks.club/newsletter.html
The Data Engineering Podcast	Podcast	Weekly	dataengineeringpodcast.com

Quick Reference — All Topics at a Glance

Month 1

Resource	Type	Time	Link / Platform
Git / GitHub	W1	Free	skills.github.com
Python	W1-2	Free+Udemy	kaggle.com/learn/python — realpython.com
SQL practice	W3	100% Free	datadriven.io — leetcode.com/studyplan/top-sql-50
SQL learning	W3	Free+Udemy	mode.com/sql-tutorial — sqlzoo.net
Cloud (AWS)	W4	Free+Udemy	skillbuilder.aws — aws.amazon.com/free

Cloud (GCP)	W4	Free+Udemy	cloudskillsboost.google — BigQuery sandbox
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Month 2

Resource	Type	Time	Link / Platform
Docker	W5	Free+Udemy	docs.docker.com/get-started
Terraform	W6	Free+Udemy	developer.hashicorp.com/terraform/tutorials
Airflow	W6	Free+Udemy	academy.astronomer.io
dbt	W7	Free+Udemy	courses.getdbt.com — github.com/dbt-labs/jaffle_shop
Iceberg / Delta Lake	W7	Free	iceberg.apache.org/docs — datalakehousehub.com
Airbyte	W8	Free (open source)	docs.airbyte.com
dlt ingestion	W8	Free (open source)	dlthub.com/docs
Spark (PySpark)	W9	Free+Udemy	community.cloud.databricks.com
Kafka	W10	Free+Udemy	developer.confluent.io/learn-kafka

Month 3

Resource	Type	Time	Link / Platform
Data Observability	W11	Free	docs.greatexpectations.io — docs.soda.io
Feature Engineering + MLflow	W12	Free+Udemy	kaggle.com/learn/feature-engineering — mlflow.org
RAG + LangChain	W12	Free+Udemy	python.langchain.com/docs — cookbook.openai.com
Vector DBs	W12	Free	pinecone.io/learn — github.com/pgvector/pgvector
Capstone UI	W13	Free	docs.streamlit.io — streamlit.io/cloud
AWS Certification	W14	Free prep + paid exam	aws.amazon.com/certification
Interview practice	W14	100% Free	datadriven.io (no paywall)